

CAPACITOR CIRCUIT — BLINK WITHOUT A CHIP!



THE FADING TORCH MYSTERY

The torch was switched off... but it kept glowing for a moment!
Where does the extra light come from?
Something inside was storing energy — and slowly letting it go.

🔊 Teacher: "Zzzzt... Flash!"

"Where does the extra light come from?"



MEET THE STORAGE TANK!

Capacitor = Temporary Electricity Water Tank

Just like a Sintex rooftop tank stores water for a short time, a capacitor stores electricity! It fills up quickly and pours it out fast — making the LED blink!

Like rooftop water tanks store water for a short time — capacitors store electricity briefly!



BATTERY VS CAPACITOR

Battery fills slow, capacitor fills & empties fast! Both work together to power your circuit!

 Battery = Deep well (Main Power Source)

 Capacitor = Small cup (Storage Tank)



LET'S PRETEND!

- Fill the cup! Slurrrp...
- Pour the water! Splash!
- Now YOU are the capacitor!
- Fill up slow... then POUR it all at once!
- Teacher cue: "Slurrrp... Splash... Blink!"

**Fast fill, fast pour = LED
blinks! ⚡**



CHARGING & DISCHARGING EXPLAINED!

A small bucket fills slowly under a leaking tap — then tips over all at once! That is exactly how a capacitor works with electricity.

Charging = filling the bucket from the tap (slowly storing energy).

Discharging = bucket pours water fast — LED blinks!



THE TINY FESTIVAL LIGHT

Meet the LED!

LED = Tiny festival light that glows with little energy. Just like mirchi lights on a rice string, LEDs are small, bright, and use very little power!



**Spot the LED glowing
in our blink circuit!**



CIRCUIT PARTS — MEET THE TEAM!

Every circuit has a team of parts working together! Let's meet each one and find out what job it does to make the LED blink!



 9V Battery
= Main power source

 Capacitor
= Storage tank

 Resistor
= Energy traffic cop

 LED
= Tiny festival light

BUILD THE BLINK CIRCUIT!

Follow each step carefully! Connect the parts in order and watch your LED blink — no chip needed!

⚡ Step 1: Connect the 9V battery snap clip to the resistor leg.

🔧 Step 2: Connect resistor to the capacitor cylinder top.

🔌 Step 3: Connect the other resistor leg to the capacitor (+) pin.

💡 Step 4: Connect capacitor to LED — watch it blink as it fills & pours!



FIX THE BLINK! QUIZ

Something is wrong with the blink circuit!
Can you guess what happened? Look at
each clue and think carefully!

Bigger tank (larger capacitor) →
LED stays on longer. Why?

No battery → No blinking, LED fades
immediately. What is missing?





SAFETY FIRST!

Before we build our blink circuit, let's make sure we stay safe and have fun! Follow these simple rules every time.

Only use 9V battery snap clips — no loose wires!

No soldering or hot tools — ever!

Ask your teacher before connecting the battery.

WHAT DID WE LEARN?

3 Big Ideas from Today!

Today we explored capacitors, batteries, and LEDs — and made a light blink without any chip!

- 🪣 Capacitor stores electricity like a water tank.
- ⚡ Battery is main power — capacitor fills and pours fast!
- 💡 LED blinks from the capacitor's quick pour of energy!





THANK YOU & KEEP EXPLORING!

Keep Tinkering, Asking & Discovering!

**You made the LED blink without a chip —
you're a Master Tinkerer!**